

CSG202: GENERATIVE AI FUNDAMENTALS

L:2 T:0 P:2 Credits:4

Course Outcomes: Through this course students should be able to

CO1 :: understand the fundamental concepts of Generative AI and its distinction from traditional machine learning.

CO2 :: explain the core components and applications of Large Language Models (LLMs).

CO3 :: apply prompt engineering techniques to effectively interact with and optimize Generative AI models, specifically within Vertex AI.

CO4 :: analyze the ethical considerations and principles of Responsible AI in the development and deployment of AI systems.

CO5 :: evaluate how Google's AI principles are implemented in practice within Google Cloud environments.

CO6 :: assess well-structured prompts to achieve desired outputs from multimodal generative models like Gemini.

Unit I

Foundations of Generative AI : Define generative AI, explain how generative AI works, describe generative AI model types, Describe generative AI applications

Unit II

Exploring Large Language Models (LLMs) : Define Large Language Models (LLMs), Describe LLM Use Cases, Explain Prompt Tuning, Describe Google's Gen AI Development tools

Unit III

Introduction to Responsible AI : Understand why Google has put AI principles in place, identify the need for a responsible AI practice within an organization,, recognize that decisions made at all stages of a project have an impact on responsible AI, recognize that organizations can design AI to fit their own business needs and values

Unit IV

Practical Prompt Design with Vertex AI : Explain the business case for responsible AI, identify ethical considerations with AI using issue spotting best practices, describe how Google developed and put their AI Principles into practice and leverage their lessons learned, Adopt a framework for how to operationalize responsible AI in your organization

Unit V

Implementing Responsible AI in Google Cloud : Overview of Google's AI Principles, Ethical AI development at Google, Responsible innovation practices, AI governance at Google, Benefits of ethical AI adoption, Building stakeholder trust, Risk management and compliance, Brand reputation and customer confidence, Responsible AI as a competitive advantage

Unit VI

Advanced Generative AI Applications : Overview of Vertex AI, Generative AI Studio, Gemini Models, Model Garden, End-to-End AI Development Workflow, Text-to-Image Generation, Image Understanding, Video Analysis, Audio Processing, Cross-Modal Reasoning, Multimodal Application Design

List of Practicals / Experiments:

List of Practicals

- Get Started with Generative AI Studio
- Generative AI with Vertex AI: Getting Started
- Generative AI with Vertex AI: Prompt Design
- Vertex AI PaLM API: Qwik Start
- Integrate Search in Applications Using Gen AI App Builder
- Prompt Design using PaLM
- Explore and Evaluate Models using Model Garden

References:

1. GOOGLE CLOUD DOCUMENTATION BY GOOGLE by GOOGLE, N A